# DANIEL MORENO CARTAGENA

**Researcher | Machine Learning Engineer | Data Scientist**

dnlmoreno.github.io www.linkedin.com/in/danielmorenoc www.github.com/dnlmoreno Google Scholar
damoreno.cartagena@gmail.com +56 9 4288 8998 Concepción, Chile

## PERFIL

I possess **experience in managing real data from diverse sources**, coupled with a **robust theoretical and practical foundation in Machine Learning and Deep Learning**. I am dedicated to expanding my knowledge and refining both technical and interpersonal skills. I am also committed, reliable, and **autonomous** team player with effective **teamwork skills**.

## EXPERIENCE

### Researcher at SPARK Program – AI & Earth Observation

**Center for Astrophysics | Harvard & Smithsonian**

September 2026 – August 2027 Boston, Massachusetts, USA

- **Developing deep learning methods for NASA TEMPO observations**, working with images and hyperspectral cubes to segment atmospheric $NO_2$ plumes, clouds, cloud shadows, and snow.

**Tools**: Python, PyTorch, HPC, Geospatial Data Pipelines.

### Machine Learning Engineer - Astronomical Alert Broker

**ALeRCE: Automatic Learning for the Rapid Classification of Events**

October 2024 - August 2026 Concepción, Chile

- **Improved production model performance** by monitoring behavior, adjusting configurations, and adapting to new data and environmental changes.

January 2024 - May 2024 Concepción, Chile

- **Deployed an astronomical transformer model** for time series and tabular data after extensive experimentation and testing.

**Tools**: Python, PyTorch, CI/CD, Docker, MLflow, Poetry, AI/HPC Supercomputing, Amazon S3, PostgreSQL, SQLite.

### Research Assistant – AI & Astrophysics

**MINGAL (Millennium Nucleus of Galaxies)**

June 2026 – August 2026 Concepción, Chile

- **Developed an end-to-end multimodal learning pipeline for JWST observations**, training dedicated self-supervised image and spectral encoders and aligning their representations through contrastive learning to learn a unified representation of high-redshift galaxies.

**Tools**: Python, PyTorch, Big Data Pipelines.

### Research Fellow – AI & Astrophysics (Two Terms)

**Harvard John A. Paulson School of Engineering (SEAS)**

June 2024 - August 2024 Boston, Massachusetts, USA

- **Adapted pre-trained vision transformers** in PyTorch and Hugging Face to classify and extract insights from large-scale astronomical time series data.

## EDUCATION

**M.Sc. in Computer Science** (2021-2023)
**University of Concepción**
**Thesis on:** Machine Learning. **GPA:** 6.4/7.0

**B.S. in Industrial Engineering** (2020-2021)
**B.Sc. in Engineering Sciences** (2016-2020)
**University of Concepción**
**Thesis on:** Operations Research. **GPA:** 6.0/7.0

## PUBLICATIONS

"***Quantifying cross-modal transfer in astronomical transient classification***", Förster F., Moreno-Cartagena, D., et al. 2026, RPS Workshop, 40th NeurIPS.

"***Generalizing Across Astronomical Surveys: Few-Shot Light Curve Classification with Astromer 2***", Donoso-Oliva C., Becker I., Protopapas, P., Cabrera-Vives, G., Cádiz-Leyton, M., Moreno-Cartagena, D., 2025, *Astronomy & Astrophysics*, 707, A170.

"***Leveraging Pre-Trained Visual Transformers for Multi-Band Light Curve Classification***", Moreno-Cartagena, D., Protopapas, P., Cabrera-Vives, G., et al. 2025, *Astronomy & Astrophysics*, 703, A41.

"***Image-Based Multi-Survey Classification of Light Curves with a Pre-Trained Vision Transformer***", Moreno-Cartagena, D., Cabrera-Vives, G., Muñoz A., et al. 2025, Machine Learning for Astrophysics Workshop, 42th ICML.

"***Astro-MoE: Mixture of Experts for Multiband Astronomical Time Series***", Cádiz-Leyton, M., Cabrera-Vives, G., Protopapas, P., Moreno-Cartagena, D., et al. 2025, Machine Learning for Astrophysics Workshop, 42th ICML.

"***Uncertainty estimation for time series classification: Exploring predictive uncertainty in transformer-based models for variable stars***", Cádiz-Leyton, M., Cabrera-Vives, G., Protopapas, P., Moreno-Cartagena, D., et al. 2024, *Astronomy & Astrophysics*, 699, A168.

"***ATAT: Astronomical Transformer for time series And Tabular data***", Cabrera-Vives, G., Moreno-Cartagena, D., Astorga, N., et al. 2024, Astronomy & Astrophysics, 689, A289.

- **Developed a multi-survey architecture** for jointly processing time series from multiple telescopes, integrating multimodal inputs (images, metadata, time series), and optimizing performance with high-performance computing infrastructure.

August 2023 - December 2023 Boston, Massachusetts, USA

- **Led experimentation on positional encodings** of a transformer model to mitigate temporal shifts across different astronomical time series datasets.
- **Contributed to the development of Astromer**, a foundational model for astronomical time series, by building and deploying deep learning pipelines in TensorFlow as part of an international team.

**Tools**: Python, TensorFlow-Keras, PyTorch, Hugging Face, Docker, MLflow, AI/HPC Supercomputing.

### Data Engineer

**Center for Data and Artificial Intelligence, University of Concepción**

May 2022 - August 2022 Concepción, Chile

- **Identified process improvement opportunities** in the paper-making machine process of 'CMPC Celulosa S.A." through data analysis.
- **Structured and preprocessed sensor data** in collaboration with domain experts for downstream analysis and visualization.

**Tools**: Python, SQL, Pandas, Industrial Process Data

### Teacher Assistant

**University of Concepción**

March 2020 - December 2023 Concepción, Chile

- **Deep Learning** taught in the **M.S. in Computer Science**.
- **Data Science** taught in the **M.S. in Computer Science**.
- **Machine Learning** taught in the **Diploma in Artificial Intelligence**.
- **Data Analytics** taught in the **M.S. in Data Science for Innovation**.
- **Corporate Finance** taught in **"Industrial Engineering"**.
- **Strategic Management** taught in **"Industrial Engineering"**.
- **Delivered week-long training sessions** on Python, data analytics, and classical machine learning to visiting student researchers at **Harvard University** and engineers at **Minera Doña Inés de Collahuasi**, Chile.

## TECHNICAL SKILLS

**Programming:** Python, R, C/C++, NetLogo.

**Machine Learning & Deep Learning:** PyTorch, TensorFlow-Keras, Scikit-learn, Hugging Face, MLflow.

**Cloud & MLOps:** AWS (EC2, S3), Docker, CI/CD, DVC, Poetry.

**Data Engineering & Analysis:** SQL, MongoDB, Pandas, Power BI, Excel.

**Web & API Development:** Astro, FastAPI, REST APIs.

**Generative AI & Automation:** ComfyUI.

**Optimization & OR:** CPLEX, DOcplex.

**Geospatial Analysis:** QGIS, GeoPandas, Shapely.

**Languages:** Spanish (Native), English (B2 - IELTS 6.0).

## CONFERENCES AND WORKSHOPS

"***Transformer-Based Astronomical Time Series Model with Uncertainty Estimation for Detecting Misclassified Instances***", Cádiz-Leyton, M., Cabrera-Vives, G., Protopapas, P., **Moreno-Cartagena, D.,** et al. 2024, LatinX in AI (LXAI) Workshop, 41st ICML, PMLR 235.

"***Positional Encodings for Light Curve Transformers: Playing with Positions and Attention***", **Moreno-Cartagena, D.**, Cabrera-Vives, G., Protopapas, P., et al. 2023, Machine Learning for Astrophysics Workshop, 40th ICML, PMLR 202.

## PUBLICATIONS IN REVIEW

"***Multimodal representation of high-redshift galaxies from JWST imaging and spectroscopy***", **Moreno-Cartagena, D.**, Cabrera-Vives, G., Herrera-Camus Rodrigo, 2026. Submitted to *Astronomy & Astrophysics*, under peer review.

## REFERENCES


**Prof. Pavlos Protopapas, Ph.D.**

@ Harvard University

pavlos@seas.harvard.edu

- Scientific Program Director at the Institute for Applied Computational Science (IACS), SEAS, Harvard University.

**Prof. Guillermo Cabrera-Vives, Ph.D.**

@ University of Concepción

guillecabrera@udec.cl

- Full Professor, Department of Computer Science, University of Concepción.
- Director of the Center for Data and Artificial Intelligence (CDIA) - UdeC.
- Director of the Automatic Learning for the Rapid Classification of Events (ALeRCE) broker.

**Prof. Francisco Förster Burón, Ph.D.**

@ University of Chile

fforster@uchile.cl

- Full Professor, Data and Artificial Intelligence Initiative (IDIA).
- Director of the Millennium Institute for Astrophysics (MAS).
- Associate Researcher, Center for Mathematical Modeling (CMM) and Center for Astrophysics and Associated Technologies (CATA).

## ICML. International Conference on Machine Learning

**at the Hawaii Convention Center, Honolulu, US**

I presented my work “Positional Encodings for Light Curve Transformers: Playing with Positions and Attention" at the 2nd ICML Workshop on Machine Learning for Astrophysics.

## MIT-Chile Research Workshop

**in Artificial Intelligence, Data Visualization, and Causal Inference**

I was part of the student cohort, engaging in public lectures, lightning talks, workshops, and poster presentations.